# Measure, Gate, Train: Budget-Aware Self-Configuring Neural Architecture Search for Unseen Data

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## 1 Motivation

The NAS Unseen-Data Challenge (AutoML 2026) evaluates a fully automated pipeline on *hidden* classification datasets under a strict wall-clock budget per dataset. All inputs arrive as image-shaped tensors, but they are semantically heterogeneous: besides natural and satellite images, several datasets are *encoded* data, such as rendered text, Sudoku boards, or chess positions, where the spatial prior of a CNN is misleading. All decisions must be made at runtime, blind.

Our starting point is the winning method of the 2025 edition, EGNAS [1], a macro–micro search (depth, width, per-layer operation) over convolutional blocks. Reproducing it surfaced two systematic weaknesses. First, rankings from short training runs are unreliable: operations that lead after a few epochs often lose after full training, so fine-grained (micro) operation search tends to *harm* final accuracy. Second, the space is confined to one architecture family: in our preliminary experiments a *fixed* 2-D dilated-convolution backbone in the style of temporal convolutional networks [3] clearly outperformed the fully searched CNN baseline on most datasets, while plain CNNs and randomly sampled wide-vocabulary architectures won precisely on the encoded datasets, mirroring the organisers' observation that random search can beat directed NAS there [2]. The leverage lies in choosing the right architecture *family* per dataset, not in fine-grained operation search.

## 2 Research Questions

**RQ1.** Can macro–micro NAS [1] be generalised across heterogeneous architecture families (plain CNNs, 2-D dilated and 1-D temporal convolution networks), making families *comparable* within a single time-budgeted search stage despite unreliable short-training rankings? **RQ2.** Can cheap, training-free measurements of the raw data decide at runtime which architectural prior fits an unseen dataset? **RQ3.** How can harmful data augmentation be detected at runtime, given that augmentation acts as regularisation (it costs accuracy early and pays late) and that, as we observe, augmentation decisions do not transfer between architecture families?

## 3 Approach

We propose a *measure–gate–train* pipeline (figure on page 2). (1) *Measure*: a training-free fingerprint of the raw tensors; the mean correlation of neighbouring pixels separates encoded from image-like data with a wide margin (Table 1). (2) *Macro stage*: for image-like data the dilated backbone is selected directly; only encoded data pays for a bake-off in which all families are trained briefly at matched wall-clock, and the leader is adopted only if it beats the robust default by more than the measured ranking noise, otherwise the pipeline *abstains* to the default, making every decision bounded-downside. (3) *Augmentation probe*: a two-arm A/B test run *after* the family choice, with arms of the chosen family, using an asymmetric rule that deviates from the default only on a clear margin (RQ3). (4) *Micro stage*: depth search anchored on an incumbent default; a deviation must win significantly *and* survive verification by re-training at a longer horizon. (5) A *random-search arm* over a wide operation vocabulary runs only on encoded data and only if budget remains [2]. (6) *Train to budget*: the winner trains on the full remaining budget with a cosine schedule annealed to zero; all stage budgets are fractions of the clock, so the pipeline degrades gracefully from many hours down to minutes (the evaluation budget is unknown).

Table 1: Training-free fingerprint on the 13 public datasets: neighbouring-pixel correlation separates the data families by 5x.

| | $\mathrm{corr}_1$ range | family |
|---|---|---|
| text / boards / puzzles (4) | −0.05 to 0.07 | encoded |
| all remaining datasets (9) | 0.38 to 1.00 | image-like |

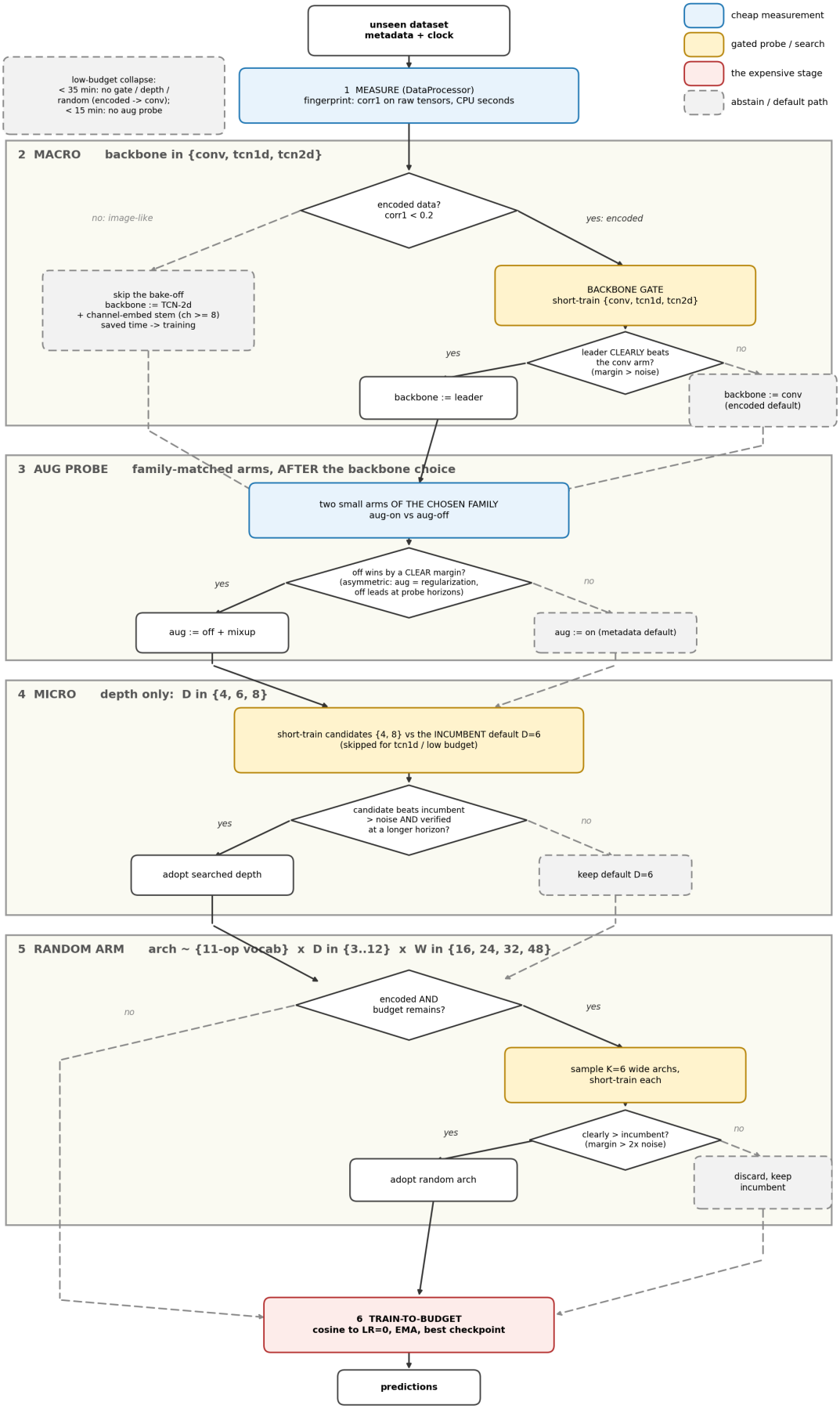
## 4 Required Steps

(1) Reproduce the EGNAS baseline and probe candidate architecture families at full budget (done, preliminary). (2) Validate each decision mechanism in isolation by replaying it against known full-budget outcomes (gate reliability, fingerprint separation, augmentation transfer). (3) Integrate the pipeline and verify all execution paths, including the low-budget collapse behaviour. (4) Full multi-seed evaluation. (5) A budget-ladder study (0.5h to 8h) measuring which decision stages pay at which budget. (6) A generalisation test on external datasets that played no role in any design choice (CIFAR-10/100, EMNIST, MedMNIST, spectrograms). (7) Submission packaging and stress tests.

## 5 Evaluation Strategy

Primary evaluation uses the 13 public competition datasets with the official adjusted score (accuracy versus a per-dataset benchmark, capped), at least three seeds for final configurations. Baselines: the reproduced EGNAS search, each fixed architecture family, and an oracle that always picks the best family, bounding the achievable gain of the routing. Every stage writes its measurements, margins, and choices to a per-dataset decision record, so gains and losses are attributable to individual decisions. Robustness is evaluated across budgets (30 minutes to 12 hours) and on the external suite, and results are compared against published per-dataset competition bests [2].

The measure-gate-train decision pipeline (run once per unseen dataset)



# References


[1] S. Siddiqui, C. Kyrkou, T. Theocharides. *Efficient Global Neural Architecture Search*. SN Computer Science 6:282, 2025.

[2] R. Geada, D. Towers, M. Forshaw, A. Atapour-Abarghouei, A. S. McGough. *Insights from the Use of Previously Unseen Neural Architecture Search Datasets*. CVPR 2024.

[3] S. Bai, J. Z. Kolter, V. Koltun. *An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling*. arXiv:1803.01271, 2018.